Google Data Analyst Case Study Track A

How can a wellness company play it smart?

By Bhagya M

High-Level Summary of the Business Task

- **Company Background:** Bellabeat is a high-tech manufacturer of health-focused products designed specifically for women (e.g., Leaf tracker, Time watch, Spring water bottle, and the Bellabeat app).
- **Business Goal:** Urška Sršen (Cofounder and CCO) wants to analyze smart device usage data from non-Bellabeat devices to uncover consumer habits.
- **Key Objective:** Use these insights to guide Bellabeat's digital marketing strategy and apply them to a specific Bellabeat product (e.g., **Bellabeat App** or **Leaf Tracker**).

Initial Data Breakdown

Running an initial exploratory script on the dataset provided key quantitative baselines:

- Sample Size: 33 unique users across 940 total daily logs.
- Daily Step Count:
    - Average: ~7,638 steps per day (below the widely recommended 10,000 steps/day health target).
    - Median: ~7,406 steps per day.
- Inactivity vs. Activity Breakdown:
    - Sedentary Minutes: Average of 991 minutes/day (~16.5 hours/day), representing the largest portion of daily time.
    - Lightly Active Minutes: Average of 193 minutes/day (~3.2 hours/day).
    - Fairly Active Minutes: Average of 13.6 minutes/day.
    - Very Active Minutes: Average of 21.2 minutes/day.

What stands out

- Steps and calories have a moderate positive relationship (correlation 0.59), so higher-movement days generally coincide with higher energy burn.

- Saturday has the highest average steps (8,131), while Sunday is lowest (6,937).
- The highest participant-wide daily average occurred on May 5 at 8,731 steps.
- 32.2% of records reached 10,000 steps; 77 records (8.2%) logged zero steps, which may include non-wear days.

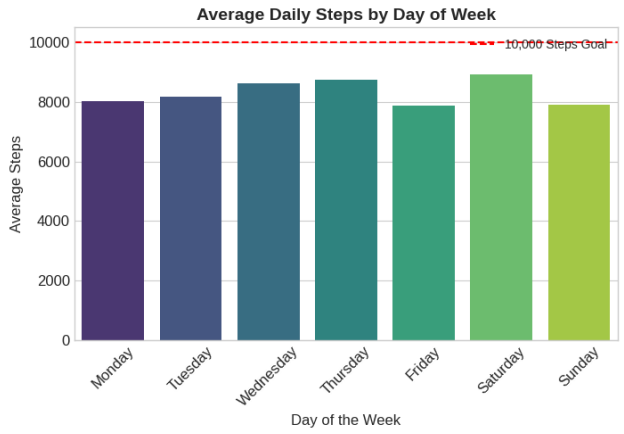
Recommended commercial actions

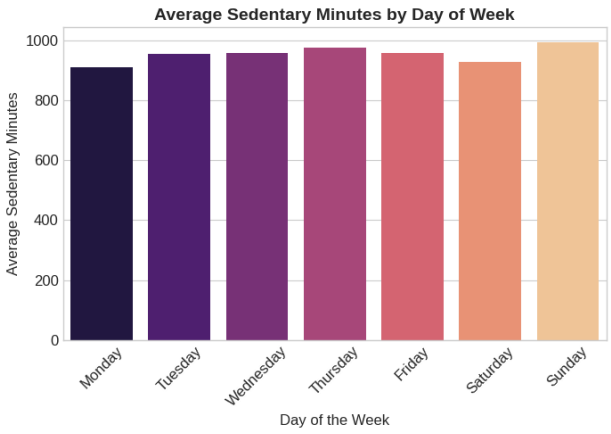
- Build a reactivation journey
    - Zero-step and low-use days indicate an adoption gap. Trigger setup help, fit/comfort tips, charging reminders, and a 7-day comeback challenge before offering upgrades.
    - Market custom goal-setting features within the Bellabeat App or membership program (e.g., "Achievable mini-goals" or personalized step recommendations for busy women).
    - Use performance messaging for active users and simple habit-building messaging for lighter users. The same product pitch should not be used for every segment.
- Use weekend challenges
    - Saturday has the strongest average step level. Launch Friday evening challenges, weekend streaks, and social referral campaigns when users are naturally more active.
- Improve the next dataset
    - Capture device model, wear location, acquisition source, tenure, campaign exposure, renewals, and purchases to measure conversion—not just activity.

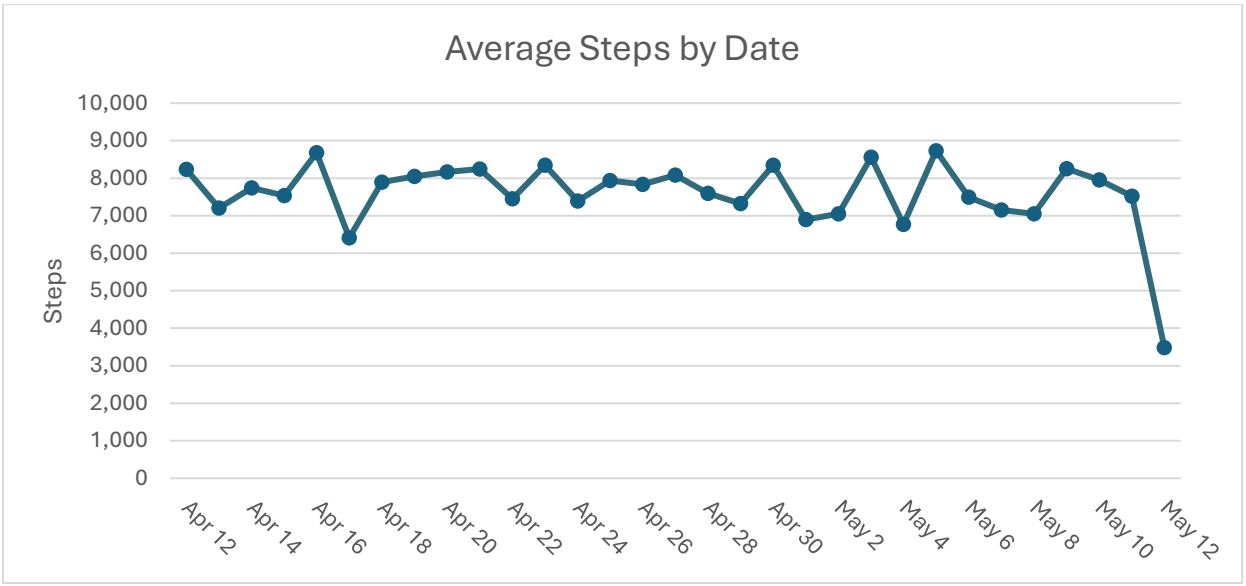
Supporting Graphs and Tables

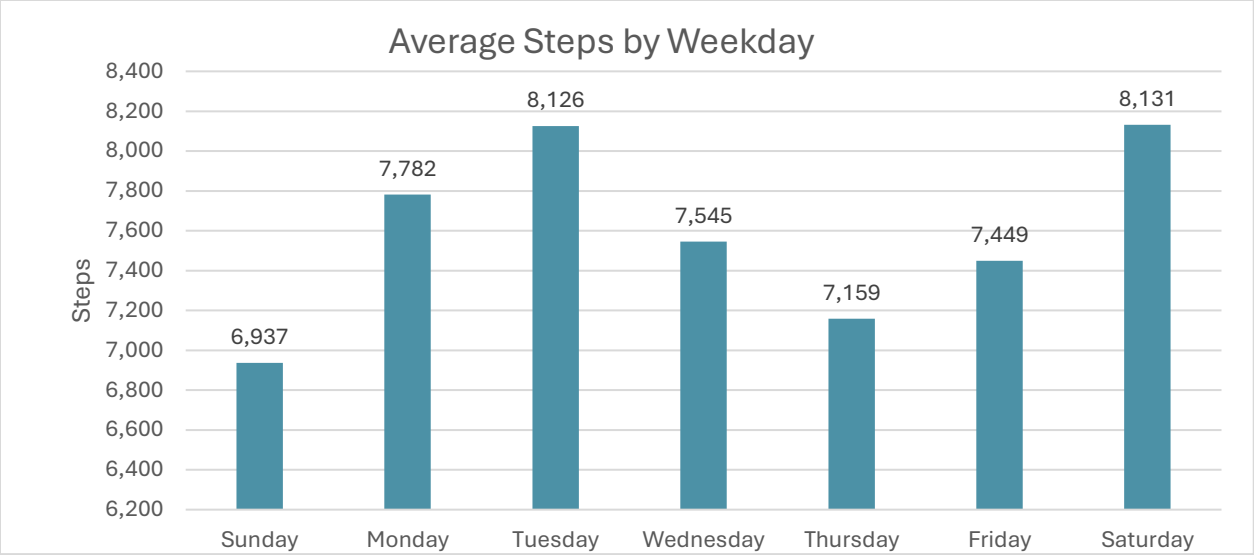
Weekly Summary Statistics:

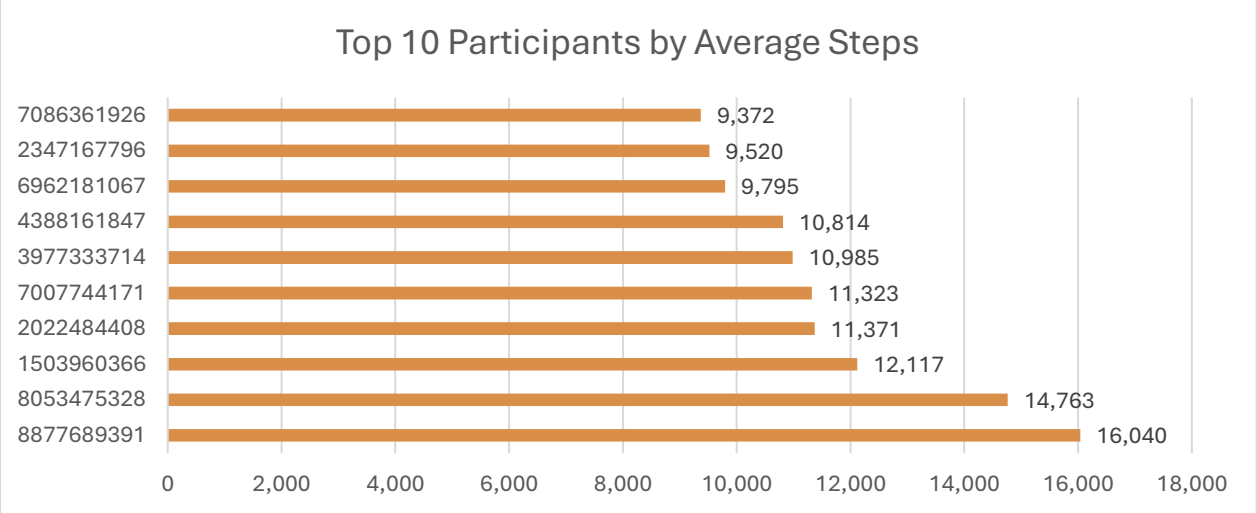
Day Of Week	Avg Steps	Avg Calories	Avg Sedentary Mins	Avg Very Active Mins	Avg Lightly Active Mins
Monday	8023.99	2214.96	911.73	23.95	197.04
Tuesday	8165.33	2357.90	956.31	22.50	205.45
Wednesday	8607.38	2390.29	959.87	25.28	209.80
Thursday	8754.56	2419.16	976.91	23.84	217.09
Friday	7858.86	2345.63	959.43	20.15	213.25
Saturday	8929.38	2413.78	929.33	24.06	222.87
Sunday	7888.31	2362.33	994.20	22.08	200.64

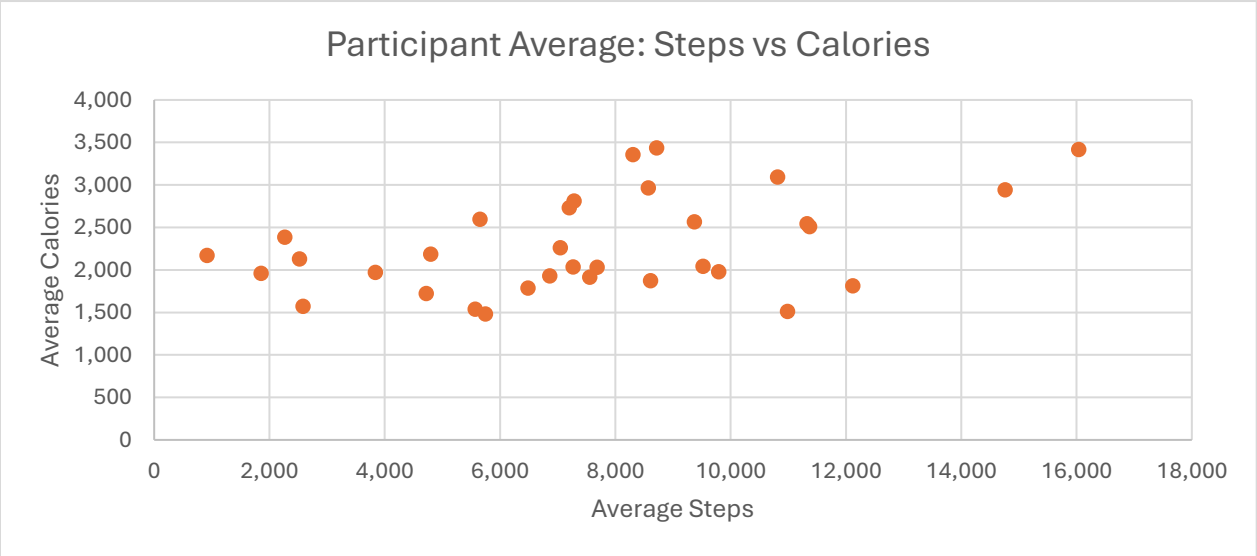







Average Steps by Weekday
Steps
8,400
8,200
8,000
7,800
7,600
7,400
7,200
7,000
6,800
6,600
6,400
6,200
6,937
7,782
8,126
7,545
7,159
7,449
8,131
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday


Top 10 Participants by Average Steps
7086361926
2347167796
6962181067
4388161847
3977333714
7007744171
2022484408
1503960366
8053475328
8877689391
9,372
9,520
9,795
10,814
10,985
11,323
11,371
12,117
14,763
16,040
0
2,000
4,000
6,000
8,000
10,000
12,000
14,000
16,000
18,000


Participant Average: Steps vs Calories
Average Calories
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
0
2,000
4,000
6,000
8,000
10,000
12,000
14,000
16,000
18,000
Average Steps

Participant	Avg Steps	Zero-Step Share	Segment
1.5E+09	12116.74	3.23%	Highly Active
1.62E+09	5743.90	0.00%	Lightly Active
1.64E+09	7282.97	0.00%	Consistent Movers
1.84E+09	2580.06	32.26%	Low Engagement
1.93E+09	916.13	45.16%	Low Engagement
2.02E+09	11370.65	0.00%	Highly Active
2.03E+09	5566.87	0.00%	Lightly Active
2.32E+09	4716.87	0.00%	Lightly Active
2.35E+09	9519.67	0.00%	Consistent Movers
2.87E+09	7555.77	0.00%	Consistent Movers
3.37E+09	6861.65	0.00%	Lightly Active
3.98E+09	10984.57	0.00%	Highly Active
4.02E+09	2267.23	45.16%	Low Engagement
4.06E+09	3838.00	25.00%	Low Engagement
4.32E+09	7268.84	0.00%	Consistent Movers
4.39E+09	10813.94	0.00%	Highly Active
4.45E+09	4796.55	0.00%	Lightly Active
4.56E+09	7685.13	0.00%	Consistent Movers
4.7E+09	8572.06	3.23%	Consistent Movers
5.55E+09	8612.58	0.00%	Consistent Movers
5.58E+09	8304.43	6.67%	Consistent Movers
6.12E+09	7046.71	17.86%	Consistent Movers
6.29E+09	5649.55	17.24%	Lightly Active
6.78E+09	2519.69	34.62%	Low Engagement
6.96E+09	9794.81	0.00%	Consistent Movers
7.01E+09	11323.42	7.69%	Highly Active
7.09E+09	9371.77	3.23%	Consistent Movers
8.05E+09	14763.29	0.00%	Highly Active
8.25E+09	6482.16	5.26%	Lightly Active
8.38E+09	8717.71	0.00%	Consistent Movers
8.58E+09	7198.52	3.23%	Consistent Movers
8.79E+09	1853.72	34.48%	Low Engagement
8.88E+09	16040.03	0.00%	Highly Active

Segment	Participants	Avg Daily Steps
Highly Active	7	12487.52
Consistent Movers	13	8225.46
Lightly Active	7	5688.222
Low Engagement	6	2329.139

